

Mellow Language Reference

Core syntax, keywords, and built-in functions

Mellow Language project

1. Overview

Mellow is an English-influenced programming language implemented in C. Function calls use angle brackets, code blocks use square brackets, and curly braces are used for array data. Arithmetic and comparison are expressed with named built-in functions rather than operators.

This reference describes the current executable core. The limitations section lists features that are not yet implemented.

2. Basic Syntax

Function calls:

```
print<"Hello, Mellow!">
add<2, 3>
print<add<2, multiply<3, 4>>>
```

Blocks:

```
if greater_than<score, 10> [
  print<"high score">
]
```

Comments use `//` for one line and `/* ... */` for multiple lines. Newlines may be used between arguments and delimiters in calls and collection literals.

3. Values and Variables

null The absence of a value. The literal is `.B null`.

number Numeric values such as `.B 42` and `.B 3.14159`.

string Text enclosed in double quotes, such as `.B "hello"`.

char A single character enclosed in single quotes. Character values use the same string representation at runtime.

bool The boolean literals `.B true` and `.B false`.

array An ordered collection written with curly braces, such as `{1, 2, 3}`. Empty `{}` is an empty array.

list An ordered collection created with `.B list<...>`.

dictionary A string-keyed mapping written with colon pairs, such as `{"name": "Ada"}`. Use `.B dict<>` for an empty dictionary.

let defines a mutable variable. Without an initializer it starts as **null**:

```
let name = "Mellow"
let unset
set<name, "Mellow Language">
```

const defines a variable that cannot be reassigned or passed as the first argument to `.B set`:

```
const PI = 3.14159
```

4. Implemented Keywords

true Creates the boolean value `true`.

false Creates the boolean value `false`.

null Creates the null value.

let Declares a mutable variable using `.B let name` or `.B let name = expression`.

const Declares an immutable variable using `.B const name` or `.B const name = expression`.

if Executes a block when an expression is truthy. Truthy values are `true`, nonzero numbers, nonempty strings, and nonempty collec-

tions.

```
if equal<name, "Mellow"> [  
    print<"matched">  
]
```

else Provides the alternative block for .B if. Chained .B else if syntax is supported. **while** Repeats a block while its condition is truthy.

```
while greater_than<count, 0> [  
    print<count>  
    set<count, dec<count>>  
]
```

loop Repeats a block indefinitely until .B break or a runtime error occurs. **break** Leaves the nearest loop. **continue** Skips to the next iteration of the nearest loop. **r return** Returns a value from a named function. With no expression it returns null. **func** Defines a named function with parameters and a square-bracket body.

```
func square<x> [  
    return multiply<x, x>  
]  
print<square<5>>
```

raise Raises a runtime exception and exits the current program with a nonzero status. The usual form is .B raise<"message">.

5. 5. Arithmetic Builtins

Arithmetic operators such as +, -, *, /, and % do not exist in Mellow. **add<a,b>** Returns the numeric sum. It also concatenates two strings or two collections of the same kind. **subtract<a,b>** Returns a minus b for numbers. **multiply<a,b>** Returns a times b for numbers. **divide<a,b>** Returns a divided by b. Division by zero is a runtime error. **modulo<a,b>** Returns the floating-point remainder of a divided by b. **power<a,b>** Returns a raised to the power b. **sqrt<a>** Returns the square root of a. **inc<a>** Returns a plus one. **dec<a>** Returns a minus one.

The retired names **sub**, **mul**, **div**, **mod**, and **pow** raise a **ValueErr** and recommend **subtract**, **multiply**, **divide**, **modulo**, and **power**, respectively. The retired names **sum**, **addition**, **minus**, **times**, **quotient**, and **remainder** also raise a **ValueErr**.

6. 6. Comparison and Logic Builtins

Comparison operators such as ==, !=, <, >, <=, and >= do not exist as operators. The following functions return booleans. **equal<a,b>** True when the values are equal, including recursive collection equality. **not_equal<a,b>** True when the values are not equal. **less_than<a,b>** True when numeric a is less than numeric b. **greater_than<a,b>** True when numeric a is greater than numeric b. **less_equal<a,b>** True when numeric a is less than or equal to numeric b. **greater_equal<a,b>** True when numeric a is greater than or equal to numeric b.

The retired names **eq**, **neq**, **lt**, **gt**, **lte**, **gte**, **less**, and **greater** raise a **ValueErr** and recommend their corresponding canonical comparison names. **and<a,b>** Returns the boolean conjunction of a and b. **or<a,b>** Returns the boolean disjunction of a and b. **not<a>** Returns the boolean negation of a. **inside<value,container>** Returns only a boolean. For strings, it checks whether the first string occurs inside the second. For arrays and lists, it checks whether an equal item exists. For dictionaries, it checks whether the string key exists. .B contains is an alias for .B inside.

7. 7. Collection and Conversion Builtins **list<items>** Creates a list from zero or more values. **len<value>** Returns the character count of a string or the item count of an array, list, or dictionary. **type<value>** Returns a string naming the runtime type: null, number, string, bool, array, list, dictionary, or instance. **to_string<value>** Converts primitive values to text. Collections currently return a collection placeholder string and are displayed directly by .B print. **print<values>** Prints each argument separated by a space, followed by a newline. **set<name,value>** Replaces the value of a mutable variable. The first argument is a variable name, not an evaluated expression. **dict<>**, **get<container, key>**, **has<container, key>**, **put<dictionary,**

key, value> Read a dictionary key or collection index, test whether it exists, or add and replace a dictionary entry. **.B dict<>** creates an empty dictionary, **.B get** returns **.B** null for a missing key, and dictionary keys must be strings. **step>range<start,end,** Creates a numeric list with an exclusive end. The step is optional and defaults to one. **for item in collection [...]** Iterates over an array or list. **input<"prompt">input<>**or Reads one terminal line as a string. An optional string prompt is printed before reading. **to_int<value>to_number<value>,to_float<value>**, Convert numeric text. **.B to_int** truncates the numeric value. **to_bool<value>** Converts booleans, nonzero numbers, or the strings true, false, 1, and 0. **lower<text>trim<text>,upper<text>**, Remove surrounding whitespace or change string case. **new>replace<text,old,** Replace every occurrence of old with new. **starts_with<text, prefix>**, **ends_with<text, suffix>** Return a boolean for prefix and suffix checks. **split<text, delimiter>**, **join<list, delimiter>** Split a string into a list or join a string collection into one string.

8. 9. Errors

One **.B** try block may have multiple consecutive **.B** catch clauses. Only the first matching handler runs. A named catch variable receives the error message; **catch<>** catches without binding; **catch<DivisionErr>** filters by type; **catch<error: DivisionErr>** filters and binds. Error types are **ValueErr**, **RecurErr**, **DivisionErr**, and **SyntaxErr**. **raise<message>** creates a runtime error. **input<"prompt">input<>**or Reads one terminal line as a string. An optional string prompt is printed before reading. **to_int<value>to_number<value>,to_float<value>**, Convert numeric text. **.B to_int** truncates the numeric value. **to_bool<value>** Converts booleans, nonzero numbers, or the strings true, false, 1, and 0. **abs<a>**, **min<a, b>**, **max<a, b>**, **clamp<a, low, high>** Common numeric helpers. **abs** returns magnitude; **min** and **max** select a number; **clamp** limits a value to a range. **round<a>floor<a>,ceil<a>**, Return the corresponding rounded numeric value. **is_null<a>**, **is_number<a>**, **is_string<a>**, **is_array<a>**, **is_list<a>**, **is_dict<a>** Return a boolean indicating whether a value has the requested runtime type.

9. 9. Modules and Imports

An import executes declarations from another Mellow file in the same runtime. Use an angle-bracket selector to import only named functions or classes; variables and top-level executable statements are ignored. Paths are relative to the process working directory and may name files in subfolders.

```
import "test/lib/greetings.mll" <welcome>
print<welcome<"Mellow">>
```

The legacy form **.B import "path.mll" <>** imports every function and class. Imported files may import other files. Import cycles are not yet detected.

10. 10. Private Methods

Inside a class, **.B private func** declares a private method. It may be called by methods of its declaring class or subclasses, including through **.B this.method<>** or **.B super.method<>**. External callers and unrelated classes are rejected. Nested classes do not automatically receive private access.

11. 8. Interpolation

Expressions inside a string using **{name}** are replaced with the current value of the named variable:

```
let user = "Ada"
print<"Hello, {user}!">
```

Interpolation currently supports variables containing strings, numbers, and booleans. Unknown names remain unchanged.

12. 9. Functions and Recursion

Named functions support parameters, local variables, return values, and recursion. Arguments are evaluated

before the function is called.

```
func factorial<n> [  
  if less_equal<n, 1> [  
    return 1  
  ]  
  return multiply<n, factorial<dec<n>>>  
]  
print<factorial<5>>
```

Functions and methods may declare a runtime-checked return type after their parameter list: -> **String**, -> **Number**, -> **Boolean**, or -> **Nothing**. **.B** Number accepts integral and fractional numbers. **.B** Nothing requires a null result. Without an annotation, return values remain dynamic.

13. 10. Object-Oriented Programming

Classes use square-bracket bodies. Fields use **.B** let and default to **.B** null without an initializer. Constructors use the method name **.B** init, and methods use angle-bracket calls.

```
class Person [  
  let name = ""  
  func init<n> [  
    set<this.name, n>  
  ]  
  func greet<> [  
    print<"Hello, ", this.name>  
  ]  
]  
let person = Person<"Ada">  
person.greet<>
```

Single inheritance uses **.B** class Student : Person [...]. Inherited fields, dynamic method lookup, overriding, and parent calls through **.B** super.method<> are supported. **.B** destroy<instance> calls an optional **.B** free<> method before releasing the instance.

Visibility defaults to **.B** public. **.B** private fields and methods are restricted to their declaring class; private top-level functions and classes are limited to their source file. **.B** static fields and methods are accessed through the class name, for example **.B** Counter.total and **.B** Counter.next<>.

14. 11. Command-Line Tools make Builds the interpreter from the project root. ./mellowfile.mll Executes a Mellow source file. file.mll./mellow--tokens Prints the token stream with source locations. ./mellow--repl Starts the interactive REPL. Type .B exit to leave it.

15. 12. Program Entry Point

If a top-level **.B** func main<> [...] exists, declarations and imports are collected and **.B** main<> runs automatically. Other top-level executable statements are skipped in this mode. Without **.B** main, the program executes top-to-bottom.

16. 13. Current Limitations

Indexing syntax, default parameters, lambdas, closures, and automatic garbage collection are not yet implemented.

